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Yannick Becker, Ralf Korn, Pascal Halffmann

Lecture 4: Insurance Applications of Continuous-time Portfolio Optimization

Ralf Korn (RPTU Kaiserslautern-Landau, Fraunhofer ITWM)

RPTU Rheinland-Pfälzische Technische Universität Kaiserslautern Landau

Fraunhofer ITWM

Aims of this Lecture:

- Show how a portfolio problem with unhedgeable risk can be solved by the stochastic control method
- Interpret it as a diffusion approximation to a non-life insurance application
- Application in creating optimal CPPI strategies

Remember the HJB-Equation approach

(8) $$0 = \sup_{u \in U} \{ g_t(t,x) + \mu(t,x,u) g_x(t,x) + {}^1\!/_2 \, \sigma^2(t,x,u) g_{xx}(t,x) \}$$

(9) $$g(T,x) = F(x) \ .$$

A three-step algorithm to solve the control problem

Step 1: (Formally) solve the optimization problem in the HJB-Equation (8) and obtain the candidate for the optimal control:

$\Rightarrow$ $u^*(t, x)$ (still depending on the unknown (!) value function and its derivatives).

Step 2: Insert $u^*(t, x)$ into the HJB-Equation (8), drop the sup-operator, solve the obtained partial differential equation with terminal condition (9) explicitly (if possible).

Step 3: Check all the assumptions made during Steps 1 and 2 (very important, often forgotten !).

A first simple insurance application: Optimal investment with insurance risk

(will be dealt with in detail in the late afternoon exercises)

Problem 1: Consider a portfolio problem with a money-market account and one stock, let $A(t)$ be the **amount of money invested in the stock** (with suitable assumptions).

$$\max_{A(.)\in P(x)} E^{0,x}\left(1-exp\left(-\lambda X^A(T)\right)\right) \text{ with}$$

$$dX^A(t) = A(t)[(b-r)dt + \sigma dW(t)] + rX^A(t)dt,\ \ X^A(0) = x$$

The corresponding HJB-Equation is given by

$$0 = \max_{A\in(-\infty,\infty)} \left\{\frac{1}{2}\sigma^2 A^2 v_{xx}(t,x) + \left(rx + A(b-r)\right)v_x(t,x) + v_t(t,x)\right\}$$
$$v(T,x) = \left(1-exp\left(-\lambda X^A(T)\right)\right)$$

Show:

The optimal amount of money kept in the stock and the wealth process are given by

$$A(t) = \frac{b-r}{\lambda\sigma^2} exp\left(-r(T-t)\right)$$ **deterministic amount of money in the stock**

$$X^A(t) = xe^{rt} + \int_0^t e^{a(t-s)}A(s)bds + \sigma\int_0^t A(s)e^{a(t-s)}dW(s)$$ **gen. Brownian motion with drift**

Note how the problem and the solution simplify for r=0. Interpret the result !

Problem 2: A portfolio problem with additional (non-hedgeable) **insurance risk**.

Approximate a risk process

$$\tilde{R}(t) = x + ct - \sum_{i=1}^{N_t} C_i \quad \text{with } C_i \text{ being i.i.d. by}$$

$$R(t) = x + at + v\widetilde{W}(t) \quad \text{with } \rho = corr\left(W, \widetilde{W}\right),\ a = c - \lambda E(C_1),\ v^2 = \lambda E\left(C_1{}^2\right)$$

For simplicity we assume *r*=0 (i.e. cash) and the stock from Problem 1 as investment opportunities. Hence, we obtain a wealth process given by

$$dX^A(t) = A(t)[bdt + \sigma dW(t)] + adt + vd\widetilde{W}(t), \quad X^A(0) = x$$

Then, the HJB-Equation to the corresponding portfolio problem is given by

$$0 = \max_A \left\{\tfrac{1}{2}(\sigma^2 A^2 + v^2 + 2\rho A\sigma v)v_{xx}(t,x,R) + (a + Ab)v_x(t,x,R) + v_t(t,x,R)\right\},$$
$$v(T,x) = 1 - exp(-\lambda x)$$

Show that the optimal strategy is given by (see (Browne 1995)):

$$A(t) \equiv A^* = \frac{b}{\lambda\sigma^2} - \rho\frac{a}{\sigma}$$ ***constant amount of money in the stock***

Analyse the influence of the correlation !

A first simple insurance application: Optimal investment with insurance risk

Some comments:

Browne (1995):

The above optimal strategy also is

- the strategy that **minimizes the probability of ruin**
- the strategy that **maximizes the probability to attain any given value** Z before ruin

Further:

If we **have *r* > 0** then

- the strategy that maximizes the exponential utility is

$$A^*(t) = \frac{b-r}{\lambda\sigma^2} e^{-r(T-t)} - \rho\frac{a}{\sigma}$$

- the strategy that minimizes the ruin probability is

$$A^*(x) = \frac{1}{b-r}\left(\sqrt{\left(rx + a - \rho\beta\left(\frac{b-r}{\sigma}\right)\right)^2 + (1-\rho^2)\beta^2\left(\frac{b-r}{\sigma}\right)^2} - (rx+a)\right)$$

A second simple insurance application: Optimal CPPI-Strategies

A guarantee component in life insurance products: The CPPI-Strategy

A promise: Full upside potential and a guaranteed lower bound

Basic idea:

- Introcude the **floor** $F(t) = Fe^{-r(T-t)}$ as the lower bound such that a guarantee value of *F* can be realized via investment in the money market account until T.
- Invest the remaining wealth $C(t) := X(t) - F(t)$, the **cushion,** in the stock.

Actually a boring strategy …

Black and Jones (1987): "Always invest a multiple $M * C(t)$ of the cushion in the stock"

Surprise: The multiple can be chosen much bigger than 1 and still the guarantee can be kept!

Theorem: (Dynamics of a CPPI)

Let $M \in IR$ and $x > e^{-rT}F$. Then, the cushion C(t) corresp. to the CPPI-strategy with constant multiplier *M* satisfies

(C1) $$dC(t) = C(t)[(Mb + (1-M)r)dt + M\sigma dW(t)],\ C(0) = x - e^{-rT}F$$

and the corresponding wealth process is given by

(C2) $$X(t) = F(t) + (x - e^{-rT}F)e^{\left(Mb+(1-M)r-0.5M^2\sigma^2\right)t+M\sigma W(t)}.$$

In particular, we have $X(T) > F$.

A guarantee component in life insurance products: The CPPI-Strategy – 2

More on the behaviour of the CPPI-Strategy:

Let X(t) be the wealth process of a CPPI-strategy with guarantee F and multiplier M. Then:

$$E\big(X(T)\big) = F + (x - Fe^{-rT})e^{(r+M(b-r))T}$$

$$Var(X(T)) = (x - Fe^{-rT})^2 e^{2(r+M(b-r))T}\left(e^{M^2\sigma^2 T} - 1\right)$$

In particular, for $b > r$ the expectation is **strictly increasing** in M, as is the variance.

Obvious question: What is the best choice of the multiplier *M* ?

Follow-up question: What is a suitable optimization problem to answer this question?

- As we have already ensured $X(T) > F$ (under suitable assumptions), we could maximize the expected utility of the overshoot of *F*, i.e. the utility from the **final value of the cushion**

New portfolio problem:

(MP) $$\max_{M(.)\in \mathrm{M}(x)} E\left(U\left(C^{(M)}(T)\right)\right)$$

with M(x) the set of all admissible multiplier processes (i.e. satisfying the conditions of portfolio processes in the usual portfolio problem) and $C^{(M)}$ denoting the cushion process corresponding to the use of the (possibly random and time-dependent) multiplier process M.

A guarantee component in life insurance products: The CPPI-Strategy – 3

Theorem (Optimality of the CPPI-Strategy)
Assume an investor with initial wealth of x > 0 wants to have a final wealth exceeding *F* with $x > e^{-rT}F$. Then, the optimal multiplier process $M^*(t)$ of the cushion process that should be invested in the stock in the sense of the portfolio problem (MP) is given by

(M_{opt}) $$M^*(t) = \begin{cases} \frac{b-r}{\sigma^2}, & \text{in case of } U(x) = \ln(x) \\ \frac{b-r}{(1-\gamma)\sigma^2}, & \text{in case of } U(x) = \frac{1}{\gamma}x^\gamma \end{cases}$$

Note:

a) In discrete time, there is a **gap risk**, i.e. the possibility that the wealth can fall below the floor between two trading times. To reduce this substantially, one often puts an upper bound on the stock position in requiring that it should not exceed the total wealth X(t).

b) The above theorem is a direct corollary from the fact that the stochastic differential equation for the cushion process looks exactly like the wealth equation in the standard portfolio problem when the multiplier process is interpreted as a portfolio process. More details can be found in e.g. Chapter 5.6 of K. (2014).

References

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